

What Moves Market Power?

Macroeconomic Determinants of Markups, Markdowns and Misallocation in Indian Manufacturing, 2008–2022*

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Abstract

What moves the market power of Indian manufacturers: the macroeconomy, the border, or neither? We study 535,215 plant-years from the Annual Survey of Industries, 2008–2022, with markups and wage markdowns estimated from production functions and misallocation measured as the dispersion of revenue productivity. Thirteen macro determinants and three policy events enter two designs fixed in a written plan before any estimate was seen: an exposure design that compares industries more and less exposed to each national shock, and an aggregate model that relates outcomes to the macro series directly. Every test is corrected for multiplicity and judged by a wild cluster bootstrap. The exposure design produces no result that meets the plan’s own standard; its single survivor, demonetisation raising markups in small-plant industries, fails its plant-level confirmation. The aggregate model finds five associations that survive correction. Trade openness and the rupee price of crude oil move inversely with markups, and both hold when the markup is measured without the first-stage correction. Wholesale inflation moves with wider wage markdowns and more monopsony incidence, as it would if nominal wages lag prices. The fifth, between private credit and misallocation, rests on a probable break in the credit series. Tariffs, on outputs or inputs, show no measurable effect on any outcome, and the plant-level exchange-rate result of an earlier version is an

*Department of Humanities and Social Sciences, Indian Institute of Technology Roorkee. This is the third paper of a series built on a plant panel constructed from the Annual Survey of Industries. The companion papers measure markups and wage markdowns (working paper 1) and misallocation under variable markups (working paper 2); this paper takes their plant-level measures as given. ASI unit records are used under licence from the Ministry of Statistics and Programme Implementation and cannot be redistributed. All aggregate panels, the tariff and macro master files, the pre-specified analysis plan and the code are available from the authors. All errors are ours.

importer-specific trend. The evidence points to price stability and openness, rather than protection, as the macro conditions that bear on market power.

JEL codes: E31, F13, F14, J42, L11, L60, O14, O25.

Keywords: markups; wage markdowns; monopsony; misallocation; inflation; trade openness; tariffs; exchange rate; demonetisation; pre-specified analysis; Indian manufacturing.

1 Introduction

Between 2008 and 2022 Indian manufacturing lived through a global financial crisis, five years of near double-digit inflation, a collapse and then a doubling of the rupee price of crude oil, a move to formal inflation targeting, the overnight withdrawal of most of the currency in circulation, a new national tax on goods and services, a pandemic lockdown, and a deliberate turn back towards tariff protection. Each of these episodes was argued over at the time in terms of prices, wages and the health of firms. Yet the question that connects them has hardly been asked for India — did any of these macroeconomic conditions change how much power manufacturers hold over the prices they charge and the wages they pay, or how efficiently labour and capital are spread across them?

The question matters for three reasons. First, market power is not a fixed trait of an industry. Markups respond to costs and competition, and wage markdowns respond to how many employers a worker can choose between and how quickly wages adjust. If the macro environment moves these wedges, then macroeconomic policy has distributional and allocative consequences that ordinary price and output statistics do not show. Second, India's trade policy changed direction after 2018, raising tariffs across a wide range of manufactured goods on the argument that domestic producers needed room to grow. Whether protection of this kind buys domestic firms pricing power, or shields workers, is an empirical question with a policy answer attached. Third, India's labour markets are thin in much of the country — a plant in a rural district may be the only large employer within reach. Where such power sits, and whether macro shocks widen it, bears directly on the wage outcomes of the manufacturing workforce.

The paper asks four research questions. The first, and the one the paper is built around, is which macroeconomic conditions move markups, wage markdowns and misallocation in Indian manufacturing. We consider thirteen national series that cover the exchange rate, the price of imported energy, wholesale and consumer inflation, interest rates, financial deepening, demand growth, the tightness of the manufacturing labour market and openness to trade. The second asks whether trade policy, through the output and input tariffs that rose after 2018, changed market power once the macro environment and the industries' own earlier trends are taken into account. The third asks whether the three large policy events of the period, demonetisation, the goods and services tax and the COVID-19 lockdown, shifted market power more in the industries most exposed to them. The fourth asks where labour-market power concentrates across states and between rural and urban India, and what it costs.

Four gaps in existing work motivate the design. The Indian evidence on market power is overwhelmingly about the trade liberalisation of the 1990s ([Krishna and Mitra, 1998](#); [Topalova and Khandelwal, 2011](#); [De Loecker et al., 2016](#)), not about the macro environment of the 2010s, which combined inflation targeting, large oil cycles and a return to

protection. The international literature on macro determinants of markups works mainly with aggregate time series for advanced economies (De Loecker et al., 2020), and the macro determinants of wage markdowns are almost unstudied, although markdowns are now measured routinely (Yeh et al., 2022; Brooks et al., 2021); the one Indian study that relates a macro variable to markdowns concerns the exchange rate and exporters (Kumar and Mallick, 2026). The Indian literature on the macro shocks of the period measures output, employment, prices and tax compliance (Mohan and Ray, 2018; Chodorow-Reich et al., 2020; Chen et al., 2023), not market power. Methodologically, a national shock takes one value per year, so the year effects that make a panel credible absorb it entirely; studies that keep the year effects cannot ask the question, and studies that drop them rarely confront how few independent observations a fifteen-year window supplies, or how many hypotheses are being tested at once. Finally, production-function markups depend on a first-stage residual (De Loecker and Warzynski, 2012; Bond et al., 2021; De Ridder et al., 2026), and a result can live entirely in that residual without anyone checking.

Our approach answers each gap in turn. We extend the plant panel of the companion papers back to 2008, giving 535,215 plant-years with markups and markdowns over fifteen years, and build a constant-basket tariff that removes the breaks created by changes in the Harmonized System. We then estimate two designs written into an analysis plan, and fixed by a cryptographic hash, before any outcome was regressed on any macro variable. The headline design interacts each national series with a predetermined industry characteristic that governs how strongly the series should bite, such as the import share of materials for the exchange rate or the oil share of inputs for the crude price, so that year effects can stay in. The secondary design relates outcomes to the macro series directly, with a common trend, and is labelled throughout as association. Every family of tests is corrected by the methods of Holm (1979) and Benjamini and Hochberg (1995), every p -value comes from a wild cluster bootstrap (Cameron et al., 2008), and every markup result is re-estimated with the markup measured without its first-stage correction.

The answer is a graded one, and we report it that way. No exposure result meets the standard the plan set in advance. One comes close — demonetisation raised the median markup of industries in which small plants produced more of the output, a result that survives correction at the industry-group level but fails the plant-level test the plan required, and disappears when the first-stage correction is removed. The aggregate model finds five associations that survive correction. Higher trade openness and a higher rupee price of crude go with lower markups, and these two are the only markup results in this paper that do not depend on the first-stage residual. Higher wholesale inflation goes with wider wage markdowns and more plants paying below the marginal revenue product of labour, which is what one would see if nominal wages lag prices. The association between private credit and misallocation cannot be interpreted, because the credit series breaks in 2016.

The trade results are clean nulls. Neither the output tariff nor the input tariff has a measurable effect on any outcome at either level of aggregation. The exchange-rate interaction that an earlier version of this paper reported as its most robust result does not survive scrutiny — at the industry level it lives entirely in the first-stage residual, and at the plant level it is explained by a steady rise in the markups of importing plants relative to non-importers that runs through the whole window and owes nothing to the exchange rate. States, space and welfare add the fourth answer. Monopsony incidence is higher among rural than urban plants in fifteen of the twenty largest manufacturing states, and states with higher output carry more misallocation.

The paper contributes in four ways. It extends the exchange-rate evidence of [Kumar and Mallick \(2026\)](#) to a broad set of macro determinants of wage markdowns, and, as far as we know, it is the first study for India to combine production-function markups and markdowns with a pre-specified, corrected test of macro determinants. It shows how an exposure design can bring national shocks into a panel with full year effects, and how far such a design can and cannot go with fifteen years of data. It corrects, with reasons, three claims made in the earlier version of this paper: that protection lowered markups, that it reduced monopsony, and that the exchange rate disciplined markups through import competition. And it turns the graded findings into specific implications for monetary, trade and labour policy, together with the statistical investments that would let the next study answer what this one cannot.

The rest of the paper runs as follows. Section 2 sets out the policy history of the period. Section 3 reviews the literature and states the gaps precisely. Section 4 describes the data and measurement, and Section 5 the designs and the inference. Section 6 reports the macro determinants, Section 7 trade and the exchange rate, and Section 8 states, space and welfare. Section 9 draws the policy implications, Section 10 discusses interpretation and limits, and Section 11 concludes. Table 1 summarises the graded findings.

2 Policy history, 2008–2022

The fifteen years studied here divide naturally into five phases, and each phase gives a different macro variable its moment. Table 2 lists the events that matter for the analysis; this section explains why each could plausibly reach market power, which is the logic behind the exposures chosen in Section 5.

Crisis and stimulus, 2008–2010. The window opens just after India cut its peak customs duty on non-agricultural goods in the 2007–08 budget, and the tariff series shows the fall — the average applied tariff dropped sharply between 2007 and 2008. The global financial crisis then hit exports and credit in the second half of 2008. The government responded with two rounds of excise duty cuts, in December 2008 and February 2009,

Table 1: The findings at a glance, graded by the standard each one meets

Grade	Finding	Evidence
Meets the pre-specified headline rule	None	Exposure design, 52 tests at NIC-3
Survives correction; association only	Trade openness and the rupee price of crude move inversely with markups WPI inflation moves with markdowns and monopsony incidence	Aggregate model, Holm .021 and .028; hold with θ^M/α^M Holm .008 and .009; fail only when lagged
Survives correction; not interpretable	Private credit and misallocation	Holm .027; credit series breaks in 2016
Survives correction; fails confirmation	Demonetisation raised markups in small-plant industries	Holm .026 at NIC-3; plant test p_{wild} .334; null with θ^M/α^M
Does not survive	Exchange rate disciplines markups of import-exposed industries Tariffs affect markups, markdowns or misallocation	Lives in the first-stage residual; plant result is an importer trend Wild p .127 to .859 except one uncorrected test
Descriptive	Rural plants face more monopsony in 15 of 20 major states; misallocation higher in larger state economies	State panel 2012–2022

and the Reserve Bank eased policy rates; the monetary and fiscal stimulus stayed in place until 2013 (Mohan and Ray, 2018). Demand recovered quickly, and by 2010 the economy was growing fast again. For manufacturers this was a period of cheap money and falling indirect taxes, followed by a sharp rebound in demand.

High inflation and a weak rupee, 2010–2014. From 2010 inflation became the dominant macro fact. Consumer inflation stayed between about nine and twelve per cent in every calendar year from 2009 to 2013, wholesale inflation was close to or above eight per cent in 2008–09, 2010–11 and 2011–12, and the Reserve Bank raised rates repeatedly. In mid-2013 the prospect of tighter monetary policy in the United States triggered capital outflows from emerging markets, and the rupee fell steeply. Mohan and Ray (2018) trace both the inflation and the external pressure of these years partly to the length of the post-crisis stimulus. A weak currency raises the rupee cost of imported inputs and the price of competing imports at the same time, so its effect on markups depends on whether a firm buys from abroad or competes with imports, which is why the paper pairs the exchange rate with the imported share of materials.

Disinflation, cheap oil and reform, 2014–2017. The second half of 2014 brought a collapse in world oil prices, and the rupee price of Brent crude in 2015–16 was less than half its 2013–14 level. Diesel prices, linked to market prices in stages from 2013, were fully deregulated in October 2014, which changed how quickly and how symmetrically oil shocks reached domestic prices (Pradeep, 2022); pass-through into Indian inflation

and industrial output had already risen once domestic fuel prices began to track world prices more often (Mandal et al., 2012). Wholesale inflation turned negative in 2015–16. Monetary policy moved to flexible inflation targeting, agreed in February 2015 and given statutory form in 2016 with a Monetary Policy Committee and a consumer-price target of four per cent; part of the disinflation that followed has been credited to favourable shocks, falling oil prices among them (Mohan and Ray, 2018). The Make in India programme began in September 2014, and the Insolvency and Bankruptcy Code was enacted in 2016.

Demonetisation and the goods and services tax, 2016–2017. On 8 November 2016 the government withdrew the legal-tender status of the 500 and 1,000 rupee notes, about 86 per cent of the currency in circulation by value (Chodorow-Reich et al., 2020). Districts more exposed to the cash shortage saw relative falls in activity and bank credit that faded within a few months (Chodorow-Reich et al., 2020), and the policy did little to meet its stated aims at a real if temporary cost in jobs and output (Lahiri, 2020). Small firms that dealt in cash bore the brunt: unregistered enterprises lost more sales than registered ones, and both stayed largely cash-dependent (Kurosaki, 2018), although the shift to electronic payment raised the sales firms reported to the tax authorities (Das et al., 2023). On 1 July 2017 the goods and services tax replaced a tangle of central and state indirect taxes with a single destination-based tax, with input credits that reward firms whose suppliers are themselves registered.

Re-protection and the pandemic, 2018–2022. The budget of 2018–19 raised customs duties on a wide range of manufactured goods, from electronics to auto parts, and further increases followed in later budgets. The turn had a precedent: after the tariff cuts of the 1990s, trade remedies such as antidumping duties and safeguards had already been used to restore protection (Bown and Tovar, 2011). The constant-basket tariff in this paper, which had drifted down from 17.3 per cent in 2012 to 15.5 per cent in 2016, returns to 17.0 per cent by 2019 and stays near that level to the end of the window. In March 2020 a national lockdown halted most manufacturing for weeks, and the migration of workers back to their villages emptied many industrial labour markets. Among micro, small and medium enterprises profits and sales fell sharply, micro-enterprises were the most fragile, and exporters were partly cushioned from the collapse in domestic demand (Chen et al., 2023). The policy response included the Atmanirbhar Bharat package and production-linked incentive schemes for selected industries, whose payouts fall mostly after our window. Inflation returned in 2021–22 as commodity prices surged, and wholesale inflation reached its highest level of the period. The four labour codes consolidating India’s labour laws were passed by Parliament in 2019 and 2020 but were not in force at any point in the window, so the period contains no change in labour regulation that could explain movements in markdowns.

Table 2: Policy timeline, 2007–2022

Date	Event	Variable in this paper
2007–08	Peak customs duty on non-agricultural goods reduced	Output tariff
Dec 2008 – Feb 2009	Excise duty cuts; monetary easing after the global crisis	Call rate, GVA growth
2010–2014	Sustained wholesale and consumer inflation; repeated rate rises	WPI and CPI inflation
Mid-2013	Capital outflows and rupee depreciation	REER, NEER
Oct 2014	Diesel prices deregulated; oil price collapse 2014–16	Brent crude in rupees
Sep 2014	Make in India launched	–
2015–2016	Flexible inflation targeting; Monetary Policy Committee	Inflation, call rate
2016	Insolvency and Bankruptcy Code	Private credit
8 Nov 2016	Demonetisation of 500 and 1,000 rupee notes	Event E1
1 Jul 2017	Goods and services tax	Event E2
2018–19 on-ward	Customs duty increases across manufactured goods	Output and input tariffs
Mar 2020	National COVID-19 lockdown	Event E3
2020	Atmanirbhar Bharat; production-linked incentive schemes	–
2021–22	Commodity-led inflation	WPI inflation

Two features of this history shape the empirical work. First, the macro series move together in ways that make their separate effects hard to identify from aggregate data alone — inflation, oil and the exchange rate all moved sharply in 2013–2016, and again in 2021–22. That is the main reason the headline design uses industry exposures rather than the series themselves. Second, the policy events were large and sudden but each happened once, so the only way to study them is to ask whether the industries most exposed to them moved more, and to treat any answer with the caution due to a single episode.

3 Literature and gaps

This section reviews four bodies of work: how market power is measured from production data, and what that measurement has shown for India; how trade and the exchange rate move markups and markdowns, again with the Indian evidence; what is known about the macro and policy shocks India went through between 2008 and 2022; and how misallocation has been studied in India. It closes by stating the gaps precisely.

3.1 Measuring market power in production data

The production approach recovers a markup from the first-order condition for a flexible input: the markup equals the output elasticity of that input divided by its share of revenue (De Loecker and Warzynski, 2012). Applied with materials as the flexible input and an estimator in the tradition of Akerberg et al. (2015), it has been used to document rising markups in the United States (De Loecker et al., 2020) and the response of Indian

markups to the tariff cuts of the 1990s (De Loecker et al., 2016). Earlier Indian work estimated price–cost margins at the industry level to ask the same question of the reforms (Balakrishnan et al., 2006). The same logic applied to labour yields a wage markdown, the gap between the marginal revenue product of labour and the wage, which is the production-side measure of monopsony power (Yeh et al., 2022; Rubens, 2023). The labour-market literature has long argued that such wedges should be common where employers are few and workers’ outside options are poor (Manning, 2003), and recent work links them to local labour-market concentration (Azar et al., 2022; Berger et al., 2022).

For India the labour side matters especially. Brooks et al. (2021) estimate that employer market power lowers labour’s share of income in Indian manufacturing by up to 13 percentage points, an effect that has fallen over time. Jayadev and Narayan (2020) document a secular decline in the labour share of India’s formal industry from 1983 that stabilised at very low levels from about 2007, and relate it to rising capital intensity, informalisation, privatisation and productivity growth in large firms. The window of this paper begins where their decline ends, so the question here is not why labour’s share fell but what moved the wedge between wages and productivity once it had.

Two critiques of the production approach matter for this paper. Bond et al. (2021) show that with revenue rather than quantity data the output elasticity is not identified in general, so markup levels should be read with care. De Ridder et al. (2026) show that markup dispersion and changes are better measured than levels. A third issue is specific to the implementation — the revenue share in the denominator is corrected by the residual of a first-stage regression of output on inputs, and any trend in that residual becomes a trend in the markup. Wage markdowns do not have this problem, because the residual cancels in the ratio of labour to materials. We return to this in Section 4, because it decides which of our markup results can be trusted.

3.2 Trade, the exchange rate and market power

The pro-competitive effect of trade is the oldest idea in this literature, and India supplies some of its best evidence, because the tariff cuts of 1991 were imposed from outside and varied across industries. Import competition compressed margins and raised productivity in the industries the reforms opened (Krishna and Mitra, 1998; Topalova and Khandelwal, 2011). Cheaper imported inputs widened the range of products Indian firms made (Goldberg et al., 2010) and raised productivity most where contracts were enforced quickly (Ahsan, 2013), and the gains differed between formal and informal producers (Nataraj, 2011). Markups moved in both directions at once: cheaper inputs raised them and import competition lowered them, with the input channel the stronger (De Loecker et al., 2016; Mukherjee and Chanda, 2021). The reforms reached labour markets too, raising the

elasticity of labour demand, and more so in states with flexible labour laws (Hasan et al., 2007). Gains from trade depend on how markups are distributed across firms (Edmond et al., 2015).

Liberalisation was not a one-way street. After the tariff cuts of the 1990s, Indian industries made increasing use of antidumping duties and safeguards to restore protection (Bown and Tovar, 2011), and tariffs everywhere respond to industry conditions as well as shaping them (Goldberg and Maggi, 1999). Both points are warnings for any design that treats the tariff rises after 2018 as exogenous. On the incidence of such rises, the evidence from the United States tariffs of the same year is that they passed fully into domestic prices, so that their cost fell on domestic buyers and importers (Amiti, Redding and Weinstein, 2019).

The exchange rate works through the same two channels with opposite signs. A stronger currency cheapens imported inputs and competing imports at once; importers and exporters adjust markups differently (Amiti et al., 2014; Berman et al., 2012), and pass-through of cost shocks into prices is incomplete in a way that moves markups (Amiti, Itskhoki and Konings, 2019; Burstein and Gopinath, 2014). Weinberger (2020) links exchange-rate shocks to markup dispersion and misallocation directly. The Indian evidence on the exchange rate is recent. Veeramani and Banerjee (2022) use plant data for 1998–2014 to show that depreciation raised gross job creation in formal manufacturing, more in states with flexible labour laws, with women’s employment used as a buffer. Closest to this paper, Kumar and Mallick (2026) estimate on Indian plant data how a depreciation moves the markups and the markdowns of exporters: it raises their markups but narrows their markdowns by more, so that their combined market power falls. Their evidence concerns exporters; ours concerns the import side and a wider set of macro shocks, and their finding that the markdown responds is one reason we study both wedges.

3.3 India’s macroeconomic and policy shocks, 2008–2022

Each macro episode of the window has an Indian literature of its own. Mohan and Ray (2018) give the narrative of monetary policy from the 2008 crisis to 2017: the joint monetary and fiscal stimulus that ran to 2013, the inflation and external instability that culminated in the taper tantrum, and the agreement of February 2015 that adopted flexible inflation targeting. On energy, pass-through of world oil prices into Indian inflation and industrial output rose once domestic fuel prices began to follow world prices more often (Mandal et al., 2012), and the market-linking of diesel prices changed the speed and the asymmetry of that pass-through (Pradeep, 2022). Energy constrains Indian manufacturers through quantity as well as price: electricity shortages reduced plants’ output (Allcott et al., 2016).

The policy events are equally well documented. Districts more exposed to demonetisation saw relative falls in economic activity and bank credit, faster adoption of electronic payment and aggregate effects that dissipated within months (Chodorow-Reich et al., 2020), and the policy had real if temporary costs while largely missing its stated objectives (Lahiri, 2020). For small firms, unregistered enterprises lost more sales than registered ones (Kurosaki, 2018), and the shift to electronic payment raised the sales firms declared for tax (Das et al., 2023). Both findings motivate the exposure we use: if cash-dependent and informal competitors bore the shock, formal plants in industries where small producers mattered had the most to gain. The 2020 lockdown cut the sales and profits of small and medium firms sharply, hit micro-enterprises hardest and spared exporters part of the fall (Chen et al., 2023). Over the longer run, manufacturing has added few jobs relative to construction and services (Erumban et al., 2019), and India’s manufacturing productivity growth has proved hard to attribute to its reforms (Bollard et al., 2013).

What this literature establishes is that each shock was large and reached manufacturing. What it does not do is measure market power. Its outcomes are output, employment, prices, credit, payments and tax compliance; none of these studies asks whether inflation, the oil price, the cash shortage or the lockdown changed the markups manufacturers charged or the markdowns they imposed on wages.

3.4 Misallocation, regulation and space in India

Misallocation is measured here as the dispersion of revenue productivity across plants in an industry (Hsieh and Klenow, 2009), a quantity that rises with any wedge that keeps marginal products apart. India is among the best-studied cases. Hsieh and Klenow (2009) find large potential gains from reallocation in Indian manufacturing. Reservation of products for small-scale firms misallocated capital and labour (García-Santana and Pijoan-Mas, 2014), and removing those reservations raised employment and output in the products affected (Martin et al., 2017). The integration of capital markets reduced misallocation (Bau and Matray, 2023), in line with the international evidence that credit constraints keep productive firms small (Midrigan and Xu, 2014), although self-financing may undo much of the damage over time (Moll, 2014). Recent work separates the part of measured dispersion due to markups from that due to other wedges (Liang, 2023; Kabiraj, 2020), and measurement error inflates measured dispersion, which complicates every comparison over time (Bils et al., 2021).

Labour regulation and geography shape where these wedges sit. States that amended the Industrial Disputes Act in a pro-worker direction saw lower investment, employment and output in registered manufacturing (Besley and Burgess, 2004), which is a reminder that policies meant to shift power towards workers can cost them jobs. Trade shocks

reached Indian regions unevenly because labour is immobile (Topalova, 2010), and barriers to competition have been linked to productivity (Sivadasan, 2009). The welfare cost of markups and markdowns is quantified by Edmond et al. (2023) and Berger et al. (2022), and the spatial concentration of industry is measured with the index of Ellison and Glaeser (1997).

3.5 The gaps this paper fills

Set against this literature, four gaps stand out. The first is substantive — the Indian markup literature stops before the macro shocks of the 2010s, and the Indian literature on those shocks measures output, employment and prices rather than market power. The one study that relates a macro variable to both markups and markdowns in India concerns the exchange rate and exporters (Kumar and Mallick, 2026); no study does so for inflation, energy prices, interest rates, credit, demand, openness or the policy events of the period. The second is a design problem. National shocks are absorbed by year effects, so a credible panel cannot estimate their average effect; shift-share designs (Goldsmith-Pinkham et al., 2020; Borusyak et al., 2022; Adão et al., 2019) offer a way forward by interacting the shock with a predetermined exposure, as Chodorow-Reich et al. (2020) do across districts for demonetisation, but they have rarely been applied to markups or markdowns. The third is statistical discipline. With fifteen years, thirteen candidate determinants, four outcomes and several levels of aggregation, a researcher who reports the significant results without correction will find some by chance; a pre-specified plan with correction is the standard remedy and is rarely used in this field. The fourth is measurement: production-function markup results that depend entirely on the first-stage residual are seldom checked, although Bond et al. (2021) and De Ridder et al. (2026) give every reason to do so. The rest of the paper is organised around closing these four gaps.

4 Data and measurement

4.1 The plant panel and the market-power measures

The plant data are the Annual Survey of Industries unit records for 2008–09 to 2022–23, linked across years by the common factory identifier. The companion paper estimates translog production functions separately for each two-digit industry on 2012–2022 and recovers, for each plant and year, the output elasticities of materials and labour, θ_{it}^M and θ_{it}^L . The markup and the wage markdown are

$$\mu_{it} = \frac{\theta_{it}^M}{\alpha_{it}^M e^{\hat{\varepsilon}_{it}}}, \quad \psi_{it} = \frac{\theta_{it}^L \alpha_{it}^M}{\theta_{it}^M \alpha_{it}^L}, \quad (1)$$

where α_{it}^M and α_{it}^L are the shares of materials and wages in revenue and $\hat{\varepsilon}_{it}$ is the residual from the first-stage regression of output on inputs. A plant with $\psi_{it} > 1$ pays a wage below the marginal revenue product of labour; we call the share of such plants in an industry-year its *monopsony incidence*. We use the repaired estimates of the companion paper, whose optimiser was corrected after the first version of this paper was written, and the outcomes are winsorised at the reporting bounds of that paper.

To reach back to 2008 we apply each industry’s estimated first-stage polynomial and translog coefficients, unchanged, to the 2008–2011 plants, after deflating their inputs and outputs with a 2004–05 base wholesale price index spliced to the 2011–12 base by commodity name. This adds 112,395 plant-years to the 2012–2022 panel, for a total of 535,215, and leaves every published value for 2012–2022 unchanged. There is no break at the join — the median markup is 1.276 in 2011 and 1.253 in 2012, and the median markdown 0.834 and 0.832. Appendix E reports the diagnostics and Appendix F the deflator splice.

A measurement fact that governs the reading of every markup result. The first stage has no year effects, so a common trend in the residual $\hat{\varepsilon}_{it}$ passes straight into μ_{it} . It does — the mean residual rises from -0.109 in 2008 to $+0.054$ in 2022, and over the same years the median markup falls from 1.359 to 1.155, while the ratio θ^M/α^M that omits the correction barely moves. The fall in the markup level is therefore a property of the first stage, not evidence of declining market power, and we do not interpret it. Year effects absorb a common trend, but not a residual that differs systematically across industries or plants. We therefore re-estimate every markup result with θ^M/α^M as the outcome, and report a markup result as interpretable only if it holds in both. The markdown ψ_{it} is free of the problem, because $\hat{\varepsilon}_{it}$ cancels in (1).

4.2 Misallocation

Misallocation is the within-industry variance of log revenue productivity, multiplied by 100, computed from the plant wedges of the companion misallocation paper. Those wedges exist only from 2012, so every misallocation result covers 2012–2022.

4.3 Industry and state panels

The plant measures are aggregated to three-digit industry groups (NIC-3) and two-digit divisions (NIC-2) by year, using medians for the markup and markdown and the share of plants with $\psi_{it} > 1$ for monopsony incidence. The NIC-3 panel has 1,020 group-years over 68 groups for the markup, markdown and monopsony, and 639 group-years over 60 groups for misallocation. The NIC-2 panel has 345 division-years over 23 divisions. The

Table 3: Macroeconomic series, 2008–2022

Series	Year	Mean	Min	Max	Source
REER, BIS broad (2020=100)	FY	96.38	86.91	102.00	BIS via FRED
NEER, BIS broad (2020=100)	FY	114.31	97.03	141.86	BIS via FRED
WPI inflation (%)	FY	4.96	−3.65	13.00	Office of Economic Adviser
CPI inflation (%)	CY	7.04	3.33	11.99	World Bank WDI
Call money rate (%)	FY	6.61	4.25	8.98	RBI via FRED
Lending rate (%)	CY	9.98	8.33	13.31	World Bank WDI
Real interest rate (%)	CY	4.16	−1.98	7.56	World Bank WDI
Private credit (% of GDP)	CY	45.16	36.89	52.39	World Bank WDI
Brent crude (Rs '000 per barrel)	FY	4.71	3.14	7.63	FRED
Real GVA growth (%)	FY	5.60	−4.23	8.96	India KLEMS
Manufacturing employment share (%)	FY	11.28	10.07	11.77	India KLEMS
Manufacturing value-added share (%)	FY	16.60	14.34	18.34	India KLEMS
Trade (% of GDP)	CY	46.88	37.76	55.79	World Bank WDI

FY: Indian fiscal year, April to March, labelled by its first calendar year. CY: calendar year, matched to the fiscal year that begins in it. Fifteen annual observations each.

state panel covers 2012–2022, because state output on a consistent base starts in 2011–12, and has 350 state-years.

4.4 Macroeconomic series

Table 3 lists the thirteen national series. Each was rebuilt from its original source and documented in the paper’s master data files. The real and nominal effective exchange rates are the Bank for International Settlements broad indices, averaged over the Indian fiscal year; the Brent crude price is converted to rupees at the monthly exchange rate and averaged over the fiscal year; wholesale inflation comes from the official index, spliced across the 2004–05 and 2011–12 bases; growth, the manufacturing employment share and the manufacturing value-added share come from the India KLEMS database; consumer inflation, interest rates, private credit and trade openness come from the World Development Indicators. Series reported by calendar year are matched to the fiscal year that begins in that calendar year, a convention we state rather than hide.

Two series need a warning before any result is read. Private credit as a share of GDP stands at 51.9 per cent in 2015 and 38.2 per cent in 2016, a fall of more than thirteen points in one year that stays in place for the rest of the window. We have found no event in Indian credit markets of that size in 2016, and we treat the step as a probable change in the source series rather than a real contraction. Any result that rests on private credit inherits it. The series published as India’s call money rate in the OECD’s main economic indicators is constant at exactly 6.00 per cent in every month from 2007 to 2011, the level at which the Reserve Bank’s Bank Rate was held in those years, while the market call rate moved widely. We keep the series because the plan named it, but it is unlikely to measure the cost of overnight money before 2012. No result involving it survives correction.

Table 4: Industry exposures, NIC-3, measured in 2008

Exposure	Pairing	Mean	S.d.	Min	Max
Imported share of materials	P1, P9	0.218	0.203	0.000	0.783
Crude and refined oil share of inputs	P2	0.069	0.148	0.000	0.855
Materials / output	P3	0.740	0.118	0.437	1.010
Emoluments / output	P4	0.086	0.055	0.008	0.341
Net fixed capital / output	P5, P6	0.358	0.197	0.076	0.956
Demand sensitivity, KLEMS 1990–2007	P7	1.815	1.198	0.527	3.775
log persons per Rs crore of output	P8, E3	1.636	0.815	-1.834	3.395
Output share of plants with <50 persons	E1, E2	0.163	0.139	0.001	0.600

4.5 Tariffs

Applied tariffs come from the World Integrated Trade Solution at the six-digit level of the Harmonized System, 105,991 product-years for 2007–2023, mapped to NIC through the official United Nations correspondence chain. The Harmonized System was revised in 2012, 2017 and 2022, and the set of tariff lines changes with each revision, so a simple average over all lines moves when the classification changes even if no duty changes. The largest such composition break is at 2016–17. We therefore build a constant-basket tariff that averages only the 4,077 lines present in every year; Appendix D shows the size of the breaks it removes and the share of identifying variation it keeps. An input tariff weights the output tariffs of supplying industries by their share in each industry’s purchases, using the 2012–13 supply-use table; it can be built for divisions but not for groups.

4.6 Industry exposures

Each macro series is paired with one industry characteristic measured in 2008, the first year of the window, so that the exposure cannot respond to the shock it is meant to measure. Table 4 lists them. The demand sensitivity is the industry’s output response to aggregate growth estimated on India KLEMS data for 1990–2007, strictly before the sample. The oil share of inputs includes both crude and refined products and comes from the supply-use table.

5 Empirical design

5.1 The exposure design

A national series M_t takes one value per year and is absorbed by year effects. It can still be studied if industries differ, in a way fixed before the sample, in how strongly M_t should

reach them. The headline design is

$$y_{jt} = \beta (M_t - \bar{M}) E_j + \vartheta \tau_{jt} + a_j + d_t + e_{jt}, \quad (2)$$

where y_{jt} is one of the four outcomes for industry j in year t , E_j the 2008 exposure, τ_{jt} the constant-basket output tariff, and a_j and d_t industry and year effects. The coefficient β measures whether the outcome moves more with M_t in more exposed industries. It is a differential effect — the average effect of M_t on all industries is absorbed by d_t and is not estimated. The tariff enters every regression, because the research objective concerns macro factors *along with* trade. The three policy events enter the same way, with M_t an indicator for the event year and E_j the small-plant output share (demonetisation and the goods and services tax) or labour intensity (the lockdown). Table A1 in Appendix A sets out the pairings and their rationale.

5.2 The aggregate model

The research proposal behind this paper asked directly whether the macro series move the outcomes, which the exposure design cannot answer. The secondary design does so at the cost of weaker identification:

$$y_{jt} = \gamma M_t + a_j + \lambda t + e_{jt}. \quad (3)$$

There are no year effects, and the common linear trend λt prevents a shared trend from being read as an effect. Each of the thirteen series enters alone. The model has fifteen independent observations of each series, so any other national change in the same years is a confounder, and we describe its results as associations throughout.

5.3 Trade designs

Trade is studied in three ways. At the industry level we estimate

$$y_{jt} = \vartheta \tau_{jt} + \kappa (REER_t - 100) s_j + a_j + d_t + e_{jt}, \quad (4)$$

with s_j the 2008 imported share of materials. At the plant level we compare plants inside the same division and year that differ in their own first-year import share s_i :

$$y_{ijt} = b (REER_t - 100) s_i + g (t - 2015) s_i + a_i + d_{jt} + e_{ijt}, \quad (5)$$

where a_i are plant effects and d_{jt} division-by-year effects, which absorb the tariff, every national shock and any industry-wide policy response. The term $g (t - 2015) s_i$ is an exposure-specific linear trend; we report the estimate with and without it. Finally, an

event study replaces the single interaction with one for every year relative to 2017, the year before the tariff rise, to test for trends before treatment.

5.4 Inference and the pre-specified plan

Standard errors are clustered by industry in (2) and (4), by division in (5), and by year in (3), where Driscoll–Kraay errors (Driscoll and Kraay, 1998) are reported alongside. Because the number of clusters ranges from 11 to 68, every decision uses the wild cluster bootstrap- t with Rademacher weights, the null imposed and 9,999 replications (Cameron et al., 2008). For the plant design we compute the bootstrap exactly with a Frisch–Waugh representation in which each replication costs as many operations as there are clusters, not plants.

The plan was written on 1 October 2026 before any outcome was regressed on any macro variable, and its hash was recorded. It fixed the pairings, the exposure year, the outcomes, the levels, the families and the rule for a headline result. A family is one outcome at one level, thirteen tests at NIC-3 and fourteen at NIC-2, where the input tariff can be built. Within each family we apply the corrections of Holm (1979) and Benjamini and Hochberg (1995). A pairing becomes a headline result only if it survives Holm at five per cent at NIC-3, has the same sign at NIC-2, and, where a plant-level exposure exists, has the same sign with a wild p below .05 in the plant design. Anything else is reported as suggestive or null. One amendment was made before any result was seen, adding the three policy events at the author’s request. Stress tests run after the results were known are labelled as such.

6 Macroeconomic determinants

6.1 The exposure design: four families, one survivor

Table 5 reports the four headline families at NIC-3. Of 52 tests, six have a wild bootstrap p below .05, against 2.6 expected if every null hypothesis were true, so the families are not pure noise; but only one survives the Holm correction. None of the markdown, monopsony or misallocation families has a survivor. The results do not support the idea that the macro series studied here move these wedges differentially in the industries most exposed to them, at least at a scale the data can detect. A null here is not proof of no effect — the misallocation family in particular has intervals wide enough to leave large effects unexcluded.

Two near misses deserve mention because they recur later. The exchange rate interacted with the imported share of materials lowers the median markup (-0.0177 , $p_{\text{wild}} .0095$), but its Holm-adjusted p is .114, and we show in Section 7 that it does not survive

the residual check. The rupee price of crude interacted with the oil share has a conventional clustered p that is essentially zero but a wild p of .177. The oil share is concentrated in one group, petroleum refining, at 0.855 against a mean of 0.069, so the variation that identifies the coefficient comes from very few clusters, which the bootstrap respects and the asymptotic formula does not. The gap between the two p -values is a reminder of why the plan judged every result by the bootstrap.

6.2 Demonetisation: a survivor that fails its confirmation

The one survivor is demonetisation. In 2016–17 the median markup rose by 0.189 more for each unit of the 2008 small-plant output share, with a 95 per cent interval of 0.085 to 0.293, a wild p of .002 and a Holm-adjusted p of .026. The sign is the same at NIC-2 (+0.422, p_{wild} .191), the estimate survives survey weighting (p_{wild} .018) and restriction to 2012–2022 (p_{wild} .007), and it fades when the event is lagged a year (+0.098, p_{wild} .0885), as a one-off shock should.

It does not meet the headline rule. The plan required the same sign, at $p_{\text{wild}} < .05$, when each plant’s own small-plant status in its first year replaces the industry exposure, and the plant estimate is -0.0015 with a wild p of .334, and .970 with survey weights. And with θ^M/α^M as the outcome the industry estimate loses all significance (p_{wild} .747), so it lives in the first-stage residual. A reading consistent with the story is available — demonetisation disrupted cash-based competitors outside the survey, and the formal plants in industries where small producers mattered gained pricing power as an industry, which industry-by-year effects in the plant design absorb by construction. But that reading was not the test the plan set, and the residual check fails independently of it. We report demonetisation as a result that cleared correction and failed confirmation, not as a finding.

6.3 The aggregate model: five associations

The aggregate model asks the proposal’s original question directly. Of the 52 estimates in Table 6, five survive the Holm correction within their outcome family, and Figure 1 plots the three that matter most as deviations from trend.

Openness and the price of oil go with lower markups. A one-point rise in trade as a share of GDP goes with a median markup lower by 0.0059 (p_{wild} .002, Holm .021), and a 1,000-rupee rise in the price of a barrel of Brent goes with a markup lower by 0.025 (p_{wild} .002, Holm .028). Both survive the residual check: with θ^M/α^M as the outcome they are about half the size, -0.0029 and -0.000012 per rupee, and still significant (p_{wild} .005 for both). They are the only markup results in this paper that do. The first fits the pro-competitive account of openness (Krishna and Mitra, 1998; Edmond et al., 2015): years

Table 5: Exposure design: the four pre-specified NIC-3 families

Macro variable	Exposure	Markup		Markdown		Monopsony		Misallocation		
		coef.	p_{wild}	coef.	p_{wild}	coef.	p_{wild}	coef.	p_{wild}	
P1	REER	imported-input share	−0.0177	.009	−0.0010	.920	+0.0003	.947	−0.496	.579
P2	Brent crude (INR)	oil-input share	+0.0002	.177	+0.0002	.167	+0.0000	.169	−0.0046	.102
P3	WPI inflation	materials share	+0.0158	.100	+0.0166	.558	−0.0005	.944	−0.647	.252
P4	CPI inflation	labour-cost share	+0.0285	.744	−0.163	.404	−0.0270	.588	−0.178	.975
P5	Call money rate	capital intensity	+0.0072	.547	−0.0132	.698	−0.0059	.659	−1.700	.358
P6	Private credit	capital intensity	+0.0046	.406	−0.0128	.103	−0.0024	.402	−0.236	.568
P7	Real GVA growth	demand sensitivity	+0.0014	.030	+0.0006	.826	+0.0009	.534	+0.131	.205
P8	Mfg. employment share	labour intensity	−0.0080	.726	−0.0462	.164	+0.0029	.806	−1.983	.249
P9	Trade openness	imported-input share	+0.0180	.016	+0.0031	.842	−0.0017	.595	−0.0185	.979
E1	Demonetisation	small-plant share	+0.189 [†]	.002	+0.134	.414	+0.0329	.532	+4.282	.581
E2	GST	small-plant share	+0.0349	.778	+0.0939	.618	−0.0596	.413	+8.325	.407
E3	COVID year	labour intensity	+0.0553	.159	+0.0870	.037	+0.0007	.963	+3.262	.067
T1	Output tariff	(direct)	−0.0020	.332	−0.0087	.041	−0.0013	.403	−0.0206	.854

Equation (2); group and year effects; constant-basket output tariff in every regression. p_{wild} : wild cluster bootstrap- t , 9,999 Rademacher draws, null imposed, clustered by group. [†]: survives the Holm correction at five per cent within its family of 13. Markup, markdown and monopsony: 1,020 group-years, 68 groups, 2008–2022. Misallocation: 639 group-years, 60 groups, 2012–2022. T1 is the tariff’s own coefficient from the same regressions. The NIC-2 families are in Table B1.

Table 6: Aggregate model: each macro series alone, NIC-3

Macro series	Markup		Markdown		Monopsony		Misallocation	
	coef.	p_{wild}	coef.	p_{wild}	coef.	p_{wild}	coef.	p_{wild}
REER	+0.0031	.565	+0.0001	.969	−0.0008	.352	−0.147	.477
NEER	+0.0024	.321	+0.0049	.083	+0.0009	.248	−0.215	.005
Brent crude (INR)	−0.000025 [†]	.002	+0.000037	.045	+0.000011	.018	+0.0003	.581
WPI inflation	−0.0032	.051	+0.0149 [†]	<.001	+0.0041 [†]	<.001	+0.0047	.981
CPI inflation	−0.0083	.113	+0.0247	.055	+0.0067	.051	+0.514	.181
Call money rate	−0.0170	.041	−0.0197	.043	−0.0041	.205	+0.558	.624
Lending rate	+0.0197	.258	−0.0067	.751	+0.0004	.918	−3.911	.281
Real interest rate	+0.0045	.036	−0.0170	.006	−0.0044	.031	−0.0619	.824
Private credit	−0.0055	.053	+0.0001	.983	−0.000089	.958	+0.386 [†]	.002
Real GVA growth	−0.0051	.460	+0.0050	.415	+0.0014	.452	+0.0151	.912
Mfg. employment share	−0.0566	.223	−0.0647	.339	−0.0146	.451	−1.136	.604
Mfg. value-added share	+0.0293	.107	−0.0227	.690	−0.0074	.626	−0.846	.614
Trade openness	−0.0059 [†]	.002	+0.0104	.031	+0.0030	.011	+0.101	.428

Equation (3): group effects and a common linear trend, no year effects. p_{wild} : wild cluster bootstrap by year (15 clusters; 11 for misallocation). [†]: survives Holm at five per cent within its outcome family of 13. Driscoll–Kraay p -values agree in every surviving case. 1,034 group-years for the markup, markdown and monopsony; 639 for misallocation. Associations, not effects.

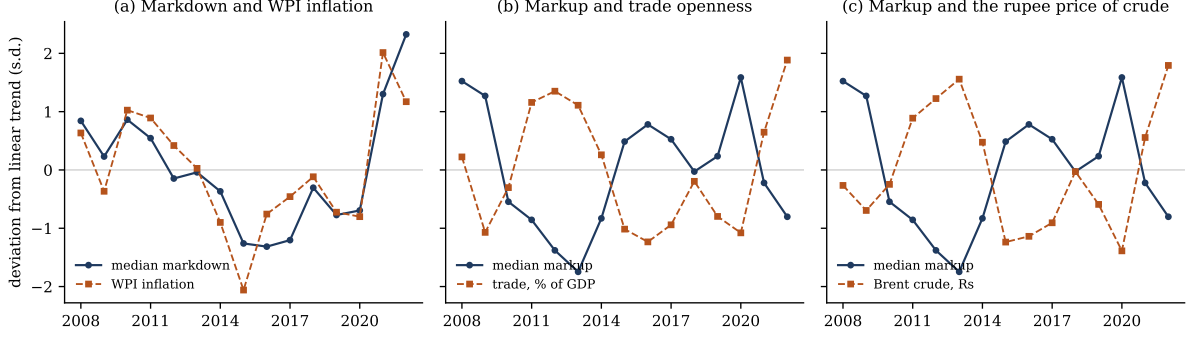


Figure 1: The three aggregate associations, as deviations from a linear trend

Mean over NIC-3 groups of the group-year median outcome, and the macro series, each detrended and divided by its standard deviation, 2008–2022. The aggregate model also removes group effects, so the figure shows the variation it uses, not the estimates themselves.

in which India traded more were years in which manufacturers charged less over cost. The second fits incomplete pass-through of cost shocks — when energy became dearer, plants absorbed part of the increase in their margins rather than passing it all into prices, and the reverse when oil was cheap (Amiti, Itskhoki and Konings, 2019; De Loecker et al., 2016). Figure 1 shows the source of the variation: the oil collapse of 2014–16, when markups were above trend, and the oil and trade surge of 2021–22, when they fell below it.

Inflation goes with wider markdowns. A one-point rise in wholesale inflation goes with a median markdown higher by 0.0149 ($p_{\text{wild}} < .001$, Holm .008) and monopsony incidence higher by 0.0041, or 0.41 percentage points of plants (Holm .009). The markdown involves no first-stage residual, so this is not a measurement echo. Its most natural reading is mechanical in the best sense — when prices rise faster than nominal wages, the wage falls relative to the marginal revenue product of labour, and by (1) the measured markdown widens. Whether that gap reflects employer power or nominal wage rigidity cannot be told apart in these data; in either case workers’ share of the value they produce shrinks during inflationary years. The association is consistent with the account of Manning (2003), in which employers with wage-setting power adjust wages only partially to changes in the value of labour. The inflationary years at the start of the window and in 2021–22 are where the markdown sits above trend in Figure 1.

Credit and misallocation cannot be read. Private credit goes with higher misallocation, +0.386 per point of GDP (Holm .027). The sign is the opposite of what the finance and misallocation literature would lead one to expect (Midrigan and Xu, 2014; Bau and Matray, 2023), and the misallocation series begins in 2012, so the step in private credit between 2015 and 2016 described in Section 4 supplies much of the identifying variation. We do not interpret this association.

Table 7: Post-hoc stress tests of the five aggregate survivors (wild p)

Association	Baseline	NIC-2	2012–22	Lag 1	Quad. trend	Drop 2020	CPI
Trade openness $\rightarrow$ markup	.002	<.001	.004	.028	<.001	.003	–
Oil price $\rightarrow$ markup	.003	<.001	.002	.021	<.001	.005	–
WPI inflation $\rightarrow$ markdown	<.001	<.001	<.001	.172	.002	<.001	.056
WPI inflation $\rightarrow$ monopsony	<.001	.001	.001	.133	.001	.001	.054
Private credit $\rightarrow$ misallocation	.002	.001	.002	.223	.101	.003	–

Wild cluster bootstrap p by year. “flip”: the sign reverses. “–”: check not applicable. Run after the pre-specified results were known.

6.4 How much the associations can bear

After the results were known we ran six further checks on the five survivors. They are not part of the plan and are labelled post hoc. Table 7 reports them. Openness and oil hold in all five checks that apply: at NIC-2, on 2012–2022, with the series lagged a year, with a quadratic trend, and dropping 2020–21. The two inflation associations hold in four of six and fail in two: with inflation lagged a year, which is expected if the link runs through the current gap between prices and wages, and when consumer replaces wholesale inflation, where they are borderline (p .056 and .054). The credit association fails with a lag and with a quadratic trend, which confirms that it should not be read.

What the aggregate associations cannot bear is a causal reading. They rest on fifteen annual observations, and two or three macro episodes supply most of the variation. Another national shift in the same years that also moved markups or wages, a change in competition policy or in the composition of the surveyed plants for instance, would produce the same pattern. The exposure design was built to address this, and its silence on these same series is informative — if openness, oil or inflation moved market power, they did not do so more strongly in the industries our exposures predict should have been most affected.

7 Trade and the exchange rate

7.1 Tariffs

Table 8 reports the industry-level trade design. The output tariff has no measurable effect on any outcome at either level once the wild bootstrap is used: p_{wild} ranges from .244 to .859, with one exception. At NIC-3 a one-point higher tariff goes with a median markdown lower by 0.0087 (p_{wild} .044), a result that stands alone and does not survive correction in its pre-specified family (Holm .492). The input tariff, which captures the cost side of protection that [Goldberg et al. \(2010\)](#) and [De Loecker et al. \(2016\)](#) emphasise, enters negatively for the markup when estimated jointly with the output tariff (–0.0300), as the cost channel predicts, but is not distinguishable from zero (p_{wild} .323; Table B3). Neither tariff moved monopsony incidence or misallocation.

Table 8: Industry-level trade design

Level	Outcome	Output tariff			REER $\times$ import share			N
		coef.	p	p_{wild}	coef.	p	p_{wild}	
NIC-2	Markup	−0.0091	.058	.244	−0.0629	.028	.007	345
	Markdown	+0.0028	.653	.686	−0.0277	.466	.534	345
	Monopsony	−0.0004	.876	.853	+0.0106	.441	.475	345
	Misallocation	−0.1799	.486	.469	−0.2211	.926	.955	253
	Markup, θ^M/α^M	−0.0016	.382	.450	−0.0068	.451	.479	345
NIC-3	Markup	−0.0020	.336	.336	−0.0669	.057	.026	1020
	Markdown	−0.0087	.004	.044	−0.0227	.804	.756	1020
	Monopsony	−0.0013	.325	.404	+0.0228	.235	.396	1020
	Misallocation	−0.0195	.859	.859	−1.9306	.417	.406	639
	Markup, θ^M/α^M	−0.0015	.173	.127	+0.0107	.603	.624	1020

Equation (4); industry and year effects; constant-basket tariff; 2008 imported share of materials. p : cluster-robust; p_{wild} : wild cluster bootstrap. 2008–2022; misallocation 2012–2022. The rows labelled θ^M/α^M repeat the markup regression with the markup not corrected by the first-stage residual.

These nulls correct the earlier version of this paper on two points. It read the negative sign of every tariff–markup coefficient as evidence that protection did not buy market power. A sign that is never significant does not support that reading, and the coefficients here are consistent with no effect. It also reported that higher tariffs lowered monopsony incidence, a “pro-worker face of protection”; with the repaired markdowns that coefficient is far from significance (p_{wild} .404 at NIC-3), and the claim is withdrawn. What can be said is narrower — the protection episode of 2018–2022 did not produce a detectable change in markups, markdowns or misallocation in the protected industries relative to others.

7.2 The exchange rate at the industry level

The real exchange rate interacted with the imported share of materials lowers the median markup, −0.0629 at NIC-2 (p_{wild} .007) and −0.0669 at NIC-3 (p_{wild} .026). Taken at face value this is import discipline — when the rupee appreciates, industries that buy more from abroad charge less over cost. But the coefficient does not survive the residual check. With θ^M/α^M as the outcome it shrinks to −0.0068 at NIC-2 (p_{wild} .479) and changes sign at NIC-3 (+0.0107, p_{wild} .624). The whole association lives in the first-stage correction, which differs systematically across industries with different import shares. It may reflect an output-price or measurement effect rather than a change in market power, and we cannot tell which.

7.3 The exchange rate at the plant level

The plant design was the earlier version’s strongest result, and it is where the extension to 2008 changes the story most. Without a trend term, a stronger real exchange rate raises

Table 9: Plant design: the exchange rate, with and without an importer-specific trend

Exposure	Weights	No trend		Exposure $\times$ trend		θ^M/α^M , no trend	
		coef.	p_{wild}	coef.	p_{wild}	coef.	p_{wild}
First-year import share	unweighted	+0.00145	.022	−0.00012	.796	−0.00138	.447
First-year import share	survey-weighted	+0.00254	<.001	+0.00037	.417	−0.00194	.236
Importer (0/1)	unweighted	+0.00122	<.001	+0.00020	.243	−0.00010	.864
Importer (0/1)	survey-weighted	+0.00167	<.001	+0.00031	.074	+0.00001	.988

Equation (5); plant and division-by-year effects; clustered by division (23); wild bootstrap computed exactly by the Frisch–Waugh representation, 9,999 draws. 525,838 plant-years, 2008–2022. The trend column was estimated after the event study below was seen and is post hoc.

the markups of plants that import more relative to plants in the same division and year that import less (Table 9). With the first-year import share the coefficient is +0.00145 (p_{wild} .022), and with survey weights +0.00254 ($p_{\text{wild}} < .001$); with a binary importer indicator, +0.00122 and +0.00167 ($p_{\text{wild}} < .001$ for both).

The event study shows why this cannot be read as an exchange-rate effect. Figure 2 plots the markup of import-exposed plants relative to others, year by year, with 2017 set to zero. The path is a steady climb from −0.046 in 2008 to +0.017 in 2020, with no break at any point at which the exchange rate moved sharply, and the joint test of no difference before 2017 rejects (p .044). The real exchange rate also trended over the window, so a regression of a trending outcome on a trending interaction finds a coefficient whether or not one causes the other. Allowing each exposure its own linear trend, the exchange-rate coefficient falls to −0.00012 (p_{wild} .796) unweighted and +0.00037 (p_{wild} .417) weighted, and the binary versions to .243 and .074. With θ^M/α^M the coefficient is never significant. We conclude that importers’ markups rose relative to non-importers’ across the whole period for reasons this design cannot identify, and that the exchange rate plays no measurable part. The trend check was run after the event study was seen, and we label it post hoc, but the event study itself is enough to rule out the earlier interpretation.

7.4 Trends before treatment

The event studies for the tariff rise of 2018 tell the same cautionary story for the industry-level outcomes (Table 10). The markup passes the test of no pre-trend at the industry level (p .637), but the markdown and monopsony incidence fail it decisively ($p < .001$), as does misallocation (p .006): the industries later protected were already diverging on these margins before protection arrived. Any industry-level association between tariffs and these outcomes would therefore be contaminated. At the plant level the markdown passes (p .723) and monopsony incidence passes (p .130), while the markup fails (p .044) for the reason shown in Figure 2.

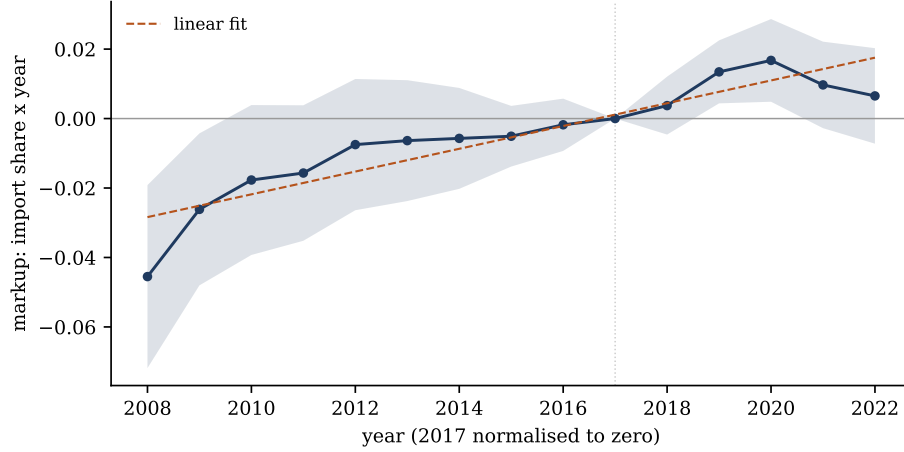


Figure 2: Markups of import-exposed plants relative to others, by year

Coefficients on the first-year import share interacted with each year, 2017 omitted; plant and division-by-year effects; 95 per cent intervals clustered by division. The dashed line is a linear fit to the coefficients.

Table 10: Joint tests of no trend before 2017

Outcome	Design	F	df	p	
Markup	industry cell	0.78	9	.637	passes
Markdown	industry cell	13.23	9	<.001	fails
Monopsony incidence	industry cell	9.88	9	<.001	fails
Misallocation	industry cell	4.40	5	.006	fails
Markup	plant	2.42	9	.044	fails
Markdown	plant	0.68	9	.723	passes
Monopsony incidence	plant	1.78	9	.130	passes

Industry cell: change in the constant-basket tariff 2017–2022 interacted with year dummies, industry and year effects. Plant: first-year import share interacted with year dummies, plant and division-by-year effects. Base year 2017. Misallocation pre-period 2012–2016.

8 States, space and welfare

The macro analysis asks what moves market power over time; this section asks where it sits. All results here cover 2012–2022, because state output on a consistent base and the misallocation wedges start then.

8.1 State output and labour-market concentration

Table 11 relates the four outcomes to the log of real state output and to the employment concentration of local labour markets, with state and year effects. Only one coefficient is distinguishable from zero — states with higher output carry more misallocation, +39.08 on the log of output (p .030). With survey weights the estimate is +31.50 and weaker (p .078). Local labour-market concentration has no measurable link to any outcome, which is notable given how central concentration is to the monopsony literature (Azar et al., 2022; Berger et al., 2022). Concentration measured at the state and three-digit industry level is a coarse proxy for the labour market a worker actually faces, and the null may reflect

Table 11: State panel: output, labour-market concentration and market power, 2012–2022

Outcome	Unweighted				Survey-weighted			
	log GSDP	p	labour HHI	p	log GSDP	p	labour HHI	p
Markup	+0.024	.693	−0.017	.910	+0.028	.628	+0.031	.847
Markdown	+0.091	.688	−0.273	.300	+0.132	.508	−0.011	.969
Monopsony	+0.035	.512	+0.158	.280	+0.055	.226	+0.241	.073
Misallocation	+39.082	.030	+33.231	.471	+31.505	.078	+48.685	.365

State and year effects; clustered by state; 350 state-years. GSDP: real gross state domestic product, 2011–12 base, as released in March 2024. The 2021–22 and 2022–23 values have since been revised by up to 12.8 per cent, mostly in small states.

that more than an absence of the mechanism. Table G1 shows the trend in concentration: the median local market became less concentrated, from 0.266 in 2012 to 0.228 in 2022, as the average number of plants per market rose from 24.3 to 29.5.

8.2 Where monopsony is: rural and urban India

The sharpest spatial fact in the data is the gap between rural and urban plants. Figure 3 maps monopsony incidence by state and the rural–urban gap, and Table G4 lists the twenty largest manufacturing states. In fifteen of them a larger share of rural plants than urban plants pays below the marginal revenue product of labour. The gap is largest in West Bengal (19.8 points) and Uttar Pradesh (19.5), and reverses most strongly in Bihar (−29.6). A rural plant draws on a labour market in which it is often the only large employer within commuting distance, which is exactly the setting in which Manning (2003) expects wage-setting power. These are raw gaps, not estimates of a mechanism — rural and urban plants differ in industry, size and age, and the comparison does not hold those fixed.

8.3 Spatial concentration and the welfare cost

Industries differ greatly in how concentrated they are across space (Table G3). Tobacco is the most concentrated, with a spatial Herfindahl index of 0.565, followed by leather and apparel; most divisions are spread across many states. The welfare cost of market power, measured as Harberger triangles with the demand elasticities of the companion misallocation paper (Table G2), shows the product-market loss rising from 6.0 per cent of sales in 2012 to 10.5 per cent in 2022, while the sales-weighted Lerner index falls from 0.160 to 0.138, and the labour-market loss stays between 2.5 and 3.3 per cent of the wage bill. Since the average Lerner index fell, the rising loss must come from the spread of markups across plants or from the year-specific demand elasticities in the calculation; we have not decomposed the two. Because the calculation uses markup levels, it inherits the first-stage trend discussed in Section 4, and its trend should be read with that in mind.

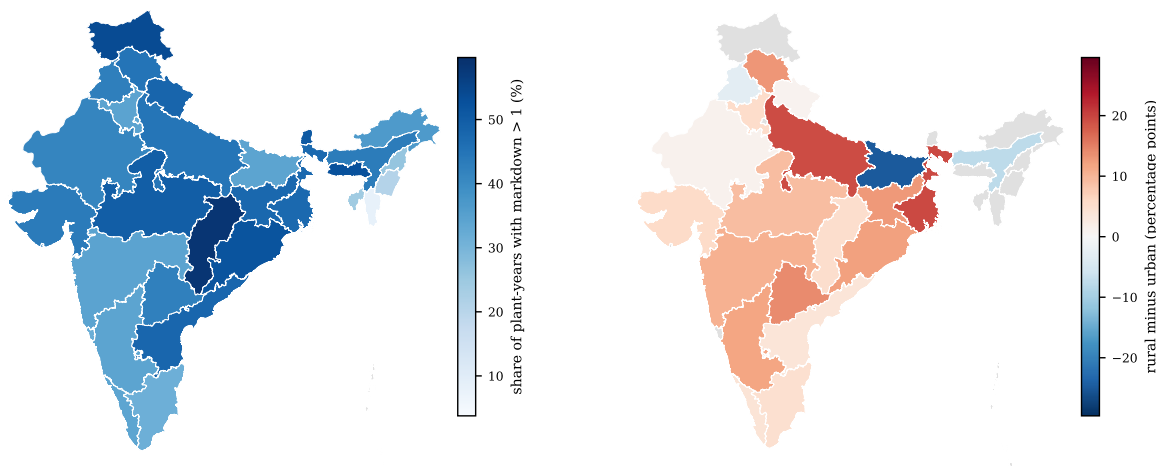


Figure 3: Monopsony incidence by state (left) and the rural minus urban gap (right), 2012–2022

Share of plant-years with $\psi_{it} > 1$. Left: states with fewer than 500 plant-years shown grey. Right: the twenty largest manufacturing states only. State boundaries follow the coding of the survey in the sample period; Ladakh, with no separate boundary in that coding, is not drawn.

9 Policy implications

The graded findings support some implications and rule out others, and it is worth being exact about which is which.

Price stability is a labour-market policy. The clearest association in the data links wholesale inflation to wider wage markdowns and to more plants paying below the value of the marginal worker. Whatever its source, the gap opens in inflationary years and narrows when inflation falls. Two implications follow. First, the inflation targeting framework agreed in 2015 and made statutory in 2016 (Mohan and Ray, 2018) has a distributional dividend in manufacturing that is not usually counted — holding inflation near target limits the erosion of workers’ share of the value they produce, a share that is already low in India’s formal industry (Jayadev and Narayan, 2020). Second, where wages are set with significant lags, as they are under periodic revision of minimum wages and multi-year settlements, faster indexation to prices would reduce the widening of markdowns in inflationary years. Both are consistent with the data; neither is proven by it, since the association is aggregate.

Openness, not protection, is associated with lower markups. Years of greater trade openness were years of lower markups, and the association survives every check we could run on it, including the one that undoes the earlier exchange-rate result. The

tariff increases after 2018 produced no detectable change in markups, markdowns or misallocation in the protected industries. So the evidence gives no support to the view that protection was needed to restore domestic firms' margins, nor to the view that it raised them at consumers' expense; and it offers no support for the claim, made in the earlier version of this paper, that protection helped workers. For a policy whose stated aim is to build domestic capacity, the absence of any measurable effect on the wedges that matter for efficiency is itself a finding. Two further cautions come from the literature rather than from our estimates. Tariffs on intermediate goods raise the costs of the downstream manufacturers who use them (Goldberg et al., 2010; De Loecker et al., 2016), and the 2018 tariffs elsewhere were paid by domestic buyers through higher prices (Amiti, Redding and Weinstein, 2019). Protection granted case by case also tends to persist once given (Bown and Tovar, 2011).

Energy prices pass into margins. Markups fell when the rupee price of crude rose, and rose when it fell. Plants absorb part of energy-cost shocks in their margins. For energy taxation and fuel-price policy this means that changes in fuel prices are shared between manufacturers and their customers rather than passed through in full, and the share borne by margins is large enough to see in aggregate data. Since the market-linking of fuel prices changed how fast and how symmetrically oil shocks reach domestic prices (Mandal et al., 2012; Pradeep, 2022), the split between margins and prices is itself a product of fuel-pricing policy.

Rural labour markets are where wage-setting power concentrates. In three quarters of the large manufacturing states, rural plants are more likely than urban ones to pay below the marginal revenue product of labour. Policies that widen workers' outside options in rural industrial areas, through transport links to larger labour markets, information on vacancies and wages, enforcement of wage floors, and skills that are portable across employers, address the margin on which the data show monopsony to be most common. Odisha, Uttar Pradesh and West Bengal, where both the level of rural incidence and the rural-urban gap are high, are natural places to start. The design of such policies matters: Indian evidence shows that regulation which raises the cost of employment without widening workers' options can reduce the employment it means to protect (Besley and Burgess, 2004), which is why the emphasis here is on outside options rather than on restrictions on employers.

Evaluating policy events needs plant-level designs. Demonetisation is the one policy event with a corrected industry-level signal, and it did not survive the plant-level test. Evaluations of large one-off events that rely on industry aggregates will find such signals by chance as often as by cause. The most persuasive evidence on demonetisation

itself comes from comparing places more and less exposed to the shock (Chodorow-Reich et al., 2020), and the same logic applies to firms. Ex post evaluations of the goods and services tax and of the production-linked incentive schemes should be designed at the plant level from the outset, with a written plan.

What the statistical system could add. Three investments would allow the next study to answer what this one cannot. The survey records the quantities of products and of imported and domestic inputs; publishing unit-value indices from them would let markups be measured without the first-stage correction that limits this paper. A consistent back-series of state output on the current base before 2011–12 would extend the state analysis to the full window. And the official trade data by six-digit product for exports as well as imports would allow import penetration to be measured directly, a better measure of competition than the tariff.

10 Interpretation and limits

Why so few results survive. Fifteen years is a short window for national shocks, and the exposure design compares industries rather than years, so it can only detect effects that differ across industries in the way the exposures predict. Many plausible macro effects are common to all manufacturing, and the design is silent on them by construction. The aggregate model is not silent, but it cannot separate one national shift from another. The two designs fail in opposite directions, which is why we report both and grade what each can bear.

The first-stage residual. The most consequential measurement result in this paper is negative — every markup result that depends on industry or plant differences in the first-stage residual disappears when the markup is measured as θ^M/α^M . This includes the exchange-rate interaction at both levels and the demonetisation result. We do not know whether the residual captures output prices, measurement error in output, or genuine variation in market power, and the companion paper’s estimates will need a first stage with year effects, or quantity data, to settle it. Until then, only markup results that hold under both measures, here openness and oil, should be cited as markup results.

Pre-trends and the meaning of the tariff nulls. The industries protected after 2018 were already diverging in markdowns, monopsony and misallocation. This is consistent with the political economy of protection (Goldberg and Maggi, 1999): tariffs were raised where industry conditions were already changing. It means that even a significant tariff coefficient for these outcomes would not have been causal, and that the nulls should be read as the absence of a detectable differential, not as proof of no effect. The equivalence

bounds of the earlier version show that the misallocation null is tight while the markdown null is not.

The dynamic evidence. The earlier version reported, from a panel vector autoregression, that tariff shocks lowered markups for one to four years. That claim is withdrawn. With the repaired markups and the longer window, the system at its published specification rejects its overidentifying restrictions (Hansen $J = 57.70$ on 32 degrees of freedom, $p .004$), and in levels it is unstable, with a largest root of 1.081. Local projections (Jordà, 2005), which need neither stability nor instruments, do not reproduce the path. Appendix C reports both.

External validity. The survey covers registered factories. Demonetisation and the lockdown hit unregistered enterprises hardest, and any reallocation of demand from them to registered plants is outside what these data can see except through its effect on the latter. The markdown is measured against materials, so a shock that moved the materials share for reasons unrelated to labour would show up in it; the inflation association survives the plan's checks but cannot exclude that channel.

11 Conclusion

We asked whether the macroeconomic conditions of 2008–2022 moved the market power of Indian manufacturers and the efficiency with which they used labour and capital. The answer depends on how much one asks of the data. Held to a standard fixed before any estimate was seen, and requiring confirmation at the plant level, no macro determinant qualifies. Held to the weaker standard of an association that survives correction for the number of hypotheses tested, four do: trade openness and the price of oil move inversely with markups, and wholesale inflation moves with wider wage markdowns and more monopsony. Trade policy, the subject of much of the period's debate, shows no measurable effect on any outcome, and the exchange-rate results that once looked robust turn out to be a measurement artefact at the industry level and a trend at the plant level.

The pattern is coherent. The macro conditions associated with market power are those that move prices and costs broadly: inflation, the price of imported energy and the openness of the economy. Instruments aimed at particular industries, such as tariffs, show no detectable effect on the wedges that matter for efficiency and wages. For policy, that points to price stability and openness as the macro conditions that bear on market power, and to rural labour markets as the place where wage-setting power is most common. For research, it points to the measurement of markups as the binding constraint: until output prices or quantities can be used directly, the markup results of any study built on revenue data, this one included, should be read with the first-stage residual in view.

Data availability

The Annual Survey of Industries unit records are licensed from the Ministry of Statistics and Programme Implementation and cannot be shared. All aggregate panels, the macro and tariff master files, the pre-specified analysis plan with its hash, the code and the logs are available from the authors and will be deposited with the replication package.

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A The pre-specified plan

The plan was written before any outcome was regressed on any macro variable and is reproduced in full in the replication package with its SHA-256 hash. Its key elements are the pairings in Table A1, the outcomes and levels in Section 4, the families and corrections in Section 5, and the three-part headline rule. The first draft omitted crude oil from the oil share and dropped two industries in matching the supply-use table; both errors were found and corrected while building the data, before any estimation. One amendment, adding events E1–E3, was made before any result was seen. Deviations: none. The stress tests of Table 7 and the trend check of Table 9 were run after results were known and are labelled post hoc wherever they appear.

B Full families and robustness

Table B1 reports the NIC-2 families, which replicate the NIC-3 families with 14 tests each; none survives Holm. Table B2 reports the pre-specified robustness checks for the markup family. Table B3 reports the input tariff.

The earlier version’s register of 72 industry-level estimates was re-run on the repaired, extended data: 10 of the 72 have a conventional p below .05, and none survives the Holm correction over the register, the same conclusion as before.

C Dynamic evidence

Table C1 reports the panel vector autoregression diagnostics (Love and Zicchino, 2006; Abrigo and Love, 2016) at NIC-3 with one lag and two to four lags of instruments. Hansen’s test rejects at the published depth of three in both transforms, and the levels

Table A1: Pre-specified pairings

	Macro variable	Exposure (2008)	Channel
P1	Real effective exchange rate	Imported share of materials	Imported-input cost and import competition
P2	Brent crude in rupees	Oil share of inputs	Energy cost
P3	WPI inflation	Materials share of output	Input-cost pass-through
P4	CPI inflation	Wage share of output	Real-wage erosion
P5	Call money rate	Capital intensity	Cost of capital
P6	Private credit	Capital intensity	Financial deepening
P7	Real GVA growth	Demand sensitivity, 1990–2007	Domestic demand
P8	Manufacturing employment share	Labour intensity	Labour-market tightness
P9	Trade openness	Imported share of materials	Openness
E1	Demonetisation, 2016–17	Small-plant output share	Cash-dependent competition
E2	GST, from 2017–18	Small-plant output share	Formalisation of demand
E3	COVID-19, 2020–21	Labour intensity	Lockdown and migrant labour
T1	Output tariff	(direct)	Protection
T2	Input tariff (NIC-2 only)	(direct)	Input-cost protection

system is unstable at every depth reported. The instrument counts are small relative to the number of panels, so the rejection is not the proliferation problem of [Roodman \(2009\)](#); the markup equation also fails the second-order serial correlation test of [Arellano and Bond \(1991\)](#) in the earlier version. Table C2 reports cumulative local projections. They need no stability condition, but the NIC-3 sample falls from 884 group-years at horizon zero to 544 at horizon five, and almost nothing is distinguishable from zero. We do not draw dynamic conclusions from either.

D Tariff construction

The tariff file holds 105,991 product-years. Each year arrives in its own revision of the Harmonized System, and the lines present differ across revisions and, to a lesser extent, across ordinary years. A simple average over all lines therefore changes when the line set changes. The constant basket keeps the 4,077 six-digit lines present in every year from 2007 to 2023, about 87 per cent of a typical year’s lines, which makes the composition effect zero at every transition by construction. The two series are correlated at above 0.99 at both levels; the basket keeps 82 per cent of the within-industry variation at NIC-2 and 94 per cent at NIC-3 after industry and year effects. The largest break it removes, at 2016–17, has a standard deviation across divisions of 1.7 percentage points, comparable to the total within-industry variation of the all-lines series. The file joins two downloads, and it was assembled only after the years they share had been compared line by line; the comparison is documented in the replication package.

Table B1: Exposure design: NIC-2 families

Macro variable	Exposure	Markup		Markdown		Monopsony		Misallocation	
		coef.	p_{wild}	coef.	p_{wild}	coef.	p_{wild}	coef.	p_{wild}
P1 REER	imported-input share	−0.0224	.367	−0.0012	.959	+0.0013	.885	−0.658	.602
P2 Brent crude (INR)	oil-input share	+0.0002	.235	+0.0002	.036	+0.0000	.140	−0.0037	.328
P3 WPI inflation	materials share	+0.0229	.264	+0.0271	.470	+0.0062	.624	−0.130	.850
P4 CPI inflation	labour-cost share	−0.280	.076	−0.180	.167	+0.0151	.799	−7.639	.021
P5 Call money rate	capital intensity	−0.0094	.690	+0.0317	.343	+0.0048	.633	−6.263	.080
P6 Private credit	capital intensity	−0.0045	.682	+0.0068	.587	+0.0009	.853	−1.448	.076
P7 Real GVA growth	demand sensitivity	+0.0018	.063	−0.0013	.206	−0.0001	.895	−0.0408	.587
P8 Mfg. employment share	labour intensity	−0.0390	.050	+0.0162	.784	+0.0150	.378	−1.677	.534
P9 Trade openness	imported-input share	+0.0342	.321	+0.0370	.070	+0.0009	.839	+0.0380	.938
E1 Demonetisation	small-plant share	+0.422	.191	+0.371	.365	−0.0221	.604	+0.575	.978
E2 GST	small-plant share	+0.263	.090	−0.262	.395	−0.207	.126	+21.583	.179
E3 COVID year	labour intensity	+0.0928	.038	+0.0772	.037	−0.0044	.807	+3.558	.232
T1 Output tariff	(direct)	−0.0090	.259	+0.0028	.693	−0.0005	.854	−0.180	.484
T2 Input tariff	(direct, joint)	−0.0300	.323	+0.0417	.369	−0.0046	.732	−2.128	.286

As Table 5, 23 divisions; 345 division-years (253 for misallocation). T2 enters jointly with the output tariff.

Table B2: Pre-specified robustness, NIC-3 markup family: coefficient (p_{wild})

	Baseline	Lag 1	Weighted	All-lines tariff	2012–2022	θ^M/α^M
P1	−0.0177 (.009)	−0.0080 (.042)	−0.0172 (.021)	−0.0176 (.010)	−0.0138 (.054)	+0.0068 (.230)
P2	+0.0002 (.177)	+0.0001 (.180)	+0.0002 (.251)	+0.0002 (.175)	+0.0002 (.015)	−0.0000 (.901)
P3	+0.0158 (.100)	+0.0101 (.350)	+0.0152 (.188)	+0.0159 (.092)	+0.0387 (<.001)	+0.0033 (.819)
P4	+0.0285 (.744)	+0.0184 (.842)	+0.0050 (.949)	+0.0289 (.736)	−0.0605 (.390)	+0.0290 (.994)
P5	+0.0072 (.547)	+0.0070 (.512)	+0.0124 (.315)	+0.0074 (.520)	+0.0124 (.450)	−0.0059 (.511)
P6	+0.0046 (.406)	+0.0042 (.355)	+0.0061 (.314)	+0.0046 (.404)	+0.0030 (.426)	−0.0011 (.769)
P7	+0.0014 (.030)	+0.0028 (<.001)	+0.0015 (.048)	+0.0014 (.038)	+0.0011 (.101)	+0.0008 (.296)
P8	−0.0080 (.726)	+0.0036 (.866)	−0.0060 (.838)	−0.0075 (.750)	+0.0024 (.909)	−0.0049 (.889)
P9	+0.0180 (.016)	+0.0133 (.014)	+0.0180 (.023)	+0.0180 (.015)	+0.0142 (.121)	−0.0019 (.669)
E1	+0.189 (.002)	+0.0977 (.088)	+0.201 (.018)	+0.190 (.001)	+0.174 (.007)	−0.0114 (.747)
E2	+0.0349 (.778)	+0.0103 (.941)	+0.0162 (.907)	+0.0328 (.797)	+0.0063 (.950)	−0.0493 (.636)
E3	+0.0553 (.159)	−0.0276 (.195)	+0.0439 (.287)	+0.0548 (.159)	+0.0486 (.163)	+0.0229 (.498)
T1	−0.0020 (.332)	−0.0020 (.333)	−0.0026 (.234)	−0.0018 (.349)	−0.0026 (.195)	−0.0015 (.130)

E The 2008–2011 extension

Table E1 reports the extended panel by year. The 2008–2011 values come from applying each division’s 2012–2022 coefficients to the earlier plants. The share of plants with $\psi > 1$ declines smoothly across the join. The mean first-stage residual rises throughout, which is the trend discussed in Section 4.

F Deflators for 2008–2011

The 2004–05 base wholesale price index was mapped to NIC divisions by commodity name, with the official commodity weights, because commodity codes mean different things in the two bases. Each series was rebased so that 2011–12 equals 100. On the nine months of overlap with the 2011–12 base, the materials deflator differs by 0.38 per cent on average and the capital deflator by 0.90 per cent; 17 of 23 divisions differ by less than three per cent. Apparel, which has no separate index before 2012, uses textiles, and other manufacturing uses the all-manufactures index (Table F1).

Table B3: Input and output tariffs entered jointly, NIC-2

Outcome	Input tariff		Output tariff	
	coef.	p_{wild}	coef.	p_{wild}
Markup	−0.0300	.323	−0.0090	.259
Markdown	+0.0417	.369	+0.0028	.693
Monopsony	−0.0046	.732	−0.0005	.854
Misallocation	−2.1278	.286	−0.1796	.484

Input tariff: output tariffs of supplying divisions weighted by the 2012–13 supply-use table. NIC 29 and 30 share one supply-use industry and therefore one input tariff.

Table C1: Panel VAR diagnostics, NIC-3, one lag

Transform	Lag	Instr. depth	Panels	J	df	p	Max. modulus
levels	1	2	59	23.68	16	.097	1.122
levels	1	3	58	74.00	32	<.001	1.081
levels	1	4	57	69.62	48	.022	1.225
demeaned	1	2	59	28.10	16	.031	1.045
demeaned	1	3	58	57.70	32	.004	0.836
demeaned	1	4	57	52.14	48	.316	1.009

G Additional spatial and welfare tables

Table C2: Local projections, NIC-3: cumulative response, coefficient (p)

Outcome	Shock	$h = 0$	$h = 1$	$h = 2$	$h = 3$
Markup	tariff	−0.0010 (.097)	−0.0007 (.367)	+0.0006 (.497)	+0.0004 (.723)
Markup	REER $\times$ exposure	−0.0303 (.091)	−0.0390 (.107)	+0.0153 (.157)	+0.0346 (.132)
Markdown	tariff	−0.0045 (.178)	−0.0049 (<.001)	+0.0008 (.503)	+0.0008 (.703)
Markdown	REER $\times$ exposure	−0.0372 (.124)	+0.0061 (.899)	+0.0299 (.190)	+0.0782 (.032)
Monopsony	tariff	−0.0002 (.615)	−0.0007 (.107)	−0.0007 (.295)	−0.0001 (.843)
Monopsony	REER $\times$ exposure	−0.0016 (.824)	+0.0071 (.218)	−0.0069 (.450)	+0.0063 (.411)
Misallocation	tariff	+0.0933 (.275)	+0.0164 (.708)	−0.1492 (.009)	+0.0017 (.980)
Misallocation	REER $\times$ exposure	+2.6578 (.240)	−2.7943 (.175)	+1.1734 (.622)	−0.5633 (.738)

Table E1: Extended plant panel by year, repaired estimates

Year	Plants	Median μ	Median ψ	Share $\psi > 1$ (%)	Mean $\hat{\varepsilon}$
2008	24,391	1.359	0.946	48.0	−0.109
2009	27,751	1.335	0.892	45.8	−0.098
2010	29,546	1.289	0.864	44.6	−0.069
2011	30,707	1.276	0.834	43.4	−0.053
2012	33,130	1.253	0.832	43.0	−0.030
2013	36,332	1.232	0.804	42.0	−0.018
2014	38,370	1.248	0.767	40.2	−0.023
2015	37,322	1.274	0.694	37.3	−0.028
2016	39,087	1.261	0.665	35.7	−0.017
2017	39,212	1.244	0.662	35.9	−0.012
2018	39,402	1.228	0.677	36.8	0.002
2019	39,738	1.217	0.631	34.7	0.022
2020	40,256	1.233	0.645	35.3	0.016
2021	40,899	1.198	0.718	38.7	0.025
2022	39,072	1.155	0.735	39.5	0.054

Table F1: Deflator splice: mean and maximum absolute gap on the overlap (per cent)

Series	2011–12 base index	Mean	Max
NIC 10	Manufacture of food products	1.65	2.52
NIC 11	Manufacture of beverages	2.14	3.75
NIC 12	Manufacture of tobacco products	2.20	4.60
NIC 13	Manufacture of textiles	1.49	2.35
NIC 14	Manufacture of wearing apparel	9.58	10.58
NIC 15	Manufacture of leather and related products	3.72	4.50
NIC 16	Manufacture of wood and of products of wood	3.83	5.36
NIC 17	Manufacture of paper and paper products	1.35	1.90
NIC 18	Printing and reproduction of recorded media	3.00	4.18
NIC 19	Mineral oils	2.51	4.95
NIC 20	Manufacture of chemicals and chemical produ	1.56	1.85
NIC 21	Manufacture of pharmaceuticals, medicinal c	1.21	1.88
NIC 22	Manufacture of rubber and plastics products	1.20	1.92
NIC 23	Manufacture of other non-metallic mineral p	1.44	2.76
NIC 24	Manufacture of basic metals	0.47	1.07
NIC 25	Manufacture of fabricated metal products, e	5.85	9.54
NIC 26	Manufacture of computer, electronic and opt	0.69	1.41
NIC 27	Manufacture of electrical equipment	1.25	2.94
NIC 28	Manufacture of machinery and equipment	0.90	1.65
NIC 29	Manufacture of motor vehicles, trailers and	2.81	4.52
NIC 30	Manufacture of other transport equipment	3.21	8.14
NIC 31	Manufacture of furniture	1.99	5.18
NIC 32	Other manufacturing	9.66	19.17
Capital	Manufacture of machinery and equipment	0.90	1.65
Materials	All commodities	0.38	0.95

Table G1: Local labour-market concentration by year

Year	Mean HHI	Median HHI	Plants per market
2012	0.374	0.266	24.3
2013	0.366	0.265	25.3
2014	0.361	0.256	26.5
2015	0.370	0.260	25.9
2016	0.356	0.259	27.8
2017	0.358	0.245	27.4
2018	0.361	0.248	27.2
2019	0.362	0.254	27.0
2020	0.351	0.237	28.3
2021	0.352	0.237	28.8
2022	0.343	0.228	29.5

Local labour market: state $\times$ NIC-3 $\times$ year. HHI of employment shares.

Table G2: Welfare cost of market power by year

Year	Lerner index	Product DWL (% of sales)	Labour DWL (% of wage bill)
2012	0.160	6.0	3.0
2013	0.161	7.8	2.9
2014	0.156	7.9	2.7
2015	0.126	7.3	2.6
2016	0.122	9.0	2.5
2017	0.133	8.3	2.7
2018	0.138	8.6	2.9
2019	0.137	12.1	2.6
2020	0.132	8.6	2.7
2021	0.145	7.4	3.1
2022	0.138	10.5	3.3

Table G3: Spatial concentration by division

NIC-2	Spatial HHI	Largest state's share
12	0.565	0.744
15	0.215	0.386
14	0.203	0.313
29	0.155	0.252
28	0.130	0.219
20	0.128	0.292
32	0.122	0.221
31	0.116	0.234
30	0.110	0.184
13	0.108	0.203
26	0.107	0.169
25	0.098	0.199
21	0.096	0.170
24	0.096	0.186
19	0.092	0.211
18	0.088	0.172
22	0.082	0.143
10	0.079	0.155
27	0.075	0.141
16	0.072	0.161
17	0.071	0.126
11	0.068	0.117
23	0.065	0.149

Table G4: Monopsony incidence, rural and urban, twenty largest manufacturing states (%)

State	Rural	Urban	Gap
West Bengal	58.9	39.1	+19.8
Uttar Pradesh	56.9	37.4	+19.5
Telangana	47.7	34.2	+13.5
Himachal Pradesh	47.4	34.9	+12.5
Jharkhand	54.7	42.4	+12.3
Odisha	57.3	45.9	+11.4
Karnataka	40.0	29.1	+10.9
Maharashtra	38.3	28.6	+9.7
Madhya Pradesh	56.2	47.9	+8.3
Gujarat	45.5	41.0	+4.4
Haryana	35.8	31.6	+4.1
Chhattisgarh	61.7	57.9	+3.8
Tamil Nadu	31.3	28.1	+3.2
Kerala	33.8	31.3	+2.5
Andhra Pradesh	48.7	46.4	+2.3
Rajasthan	40.3	40.7	−0.5
Uttarakhand	47.8	48.8	−0.9
Punjab	38.7	44.0	−5.3
Assam	40.3	50.9	−10.6
Bihar	21.0	50.6	−29.6